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Octubre 2018 | Caracas, Venezuela

Estrategia para Despliegue de Contenedores

Arquitectura de Soluciones

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VMWare Overview of Containers

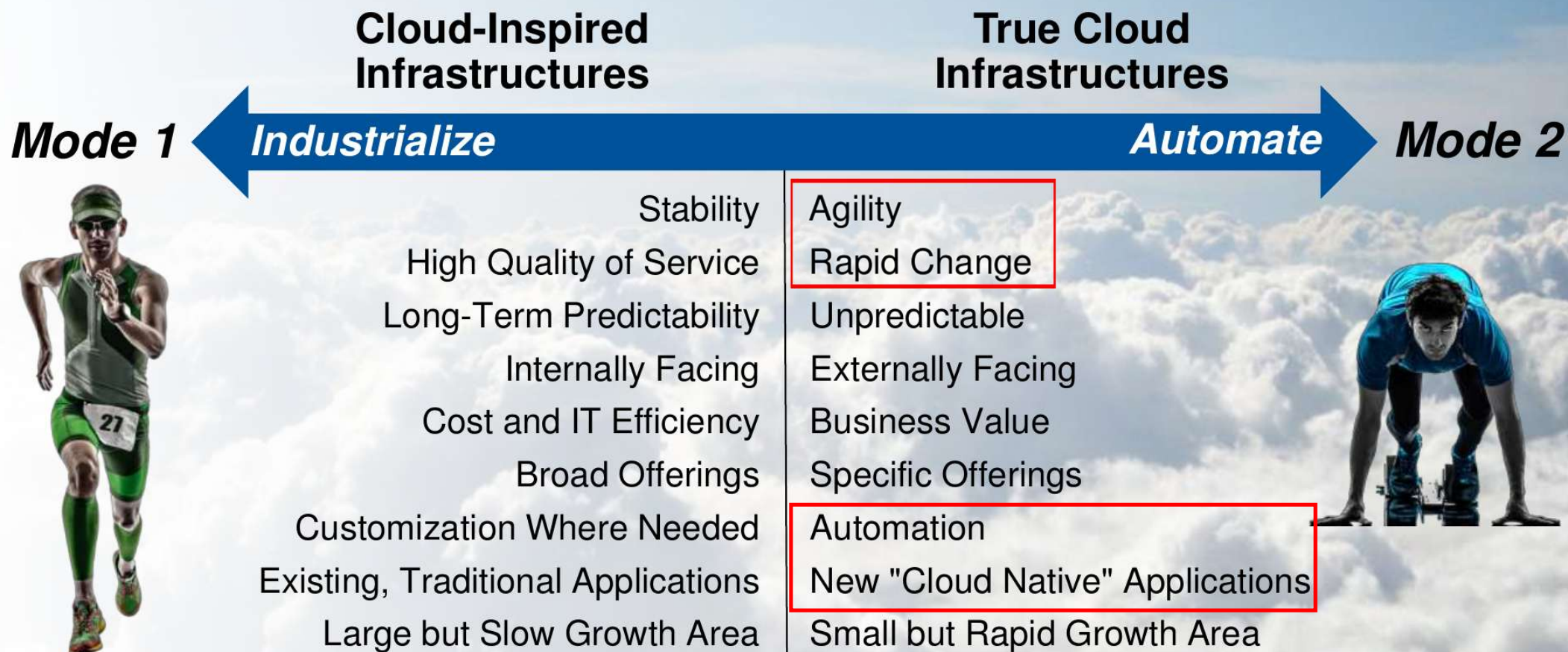
<https://www.youtube.com/watch?v=gFozhTXOx18>

Contenedores

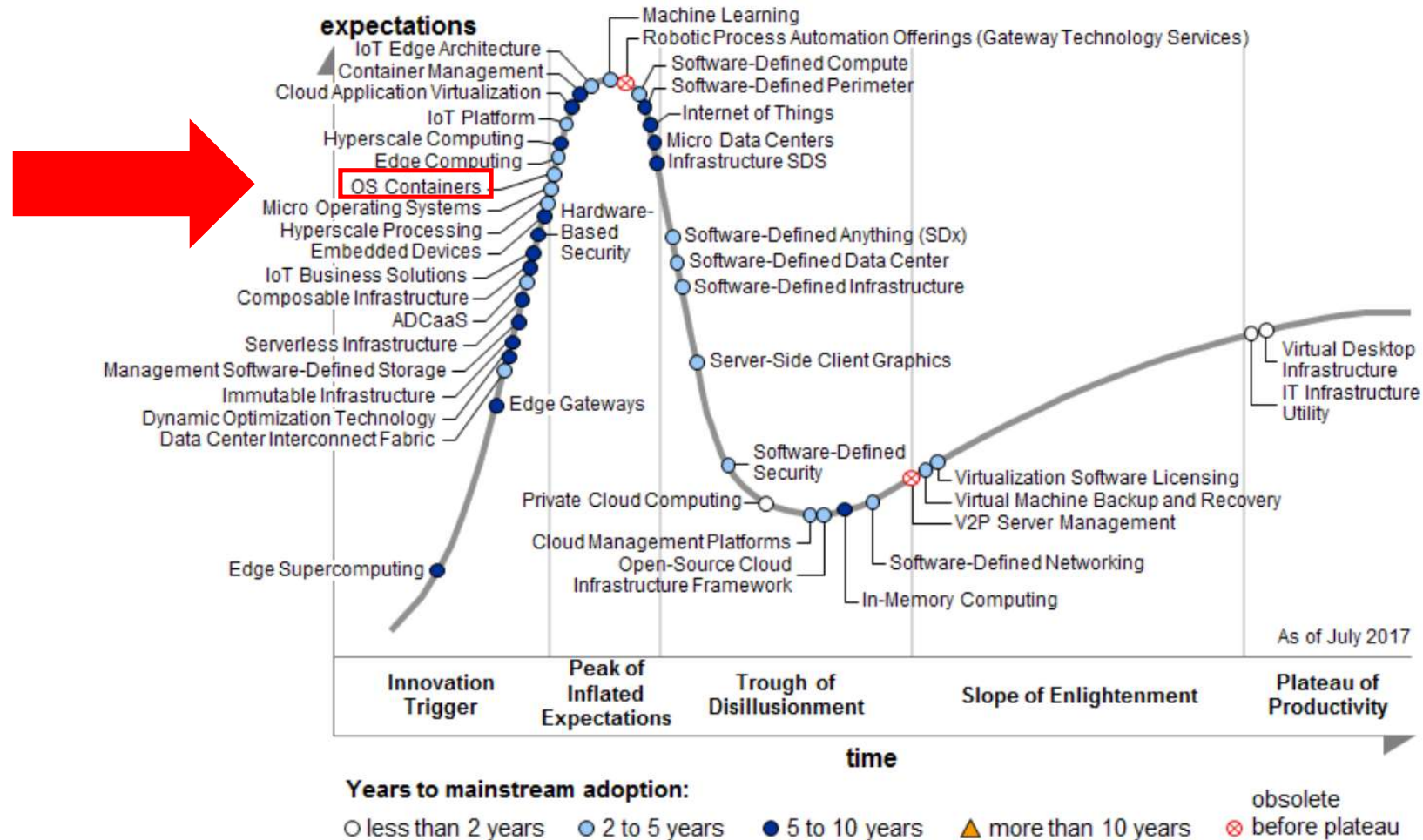
1. ¿Por qué tanto escándalo?
2. ¿Qué son los contenedores?
3. ¿Cuál es la estrategia?
4. ¿Cuál es la arquitectura?

Contenedores

- 1. ¿Por qué tanto escándalo?**
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Infrastructure Strategies' Hype Cycle 2017

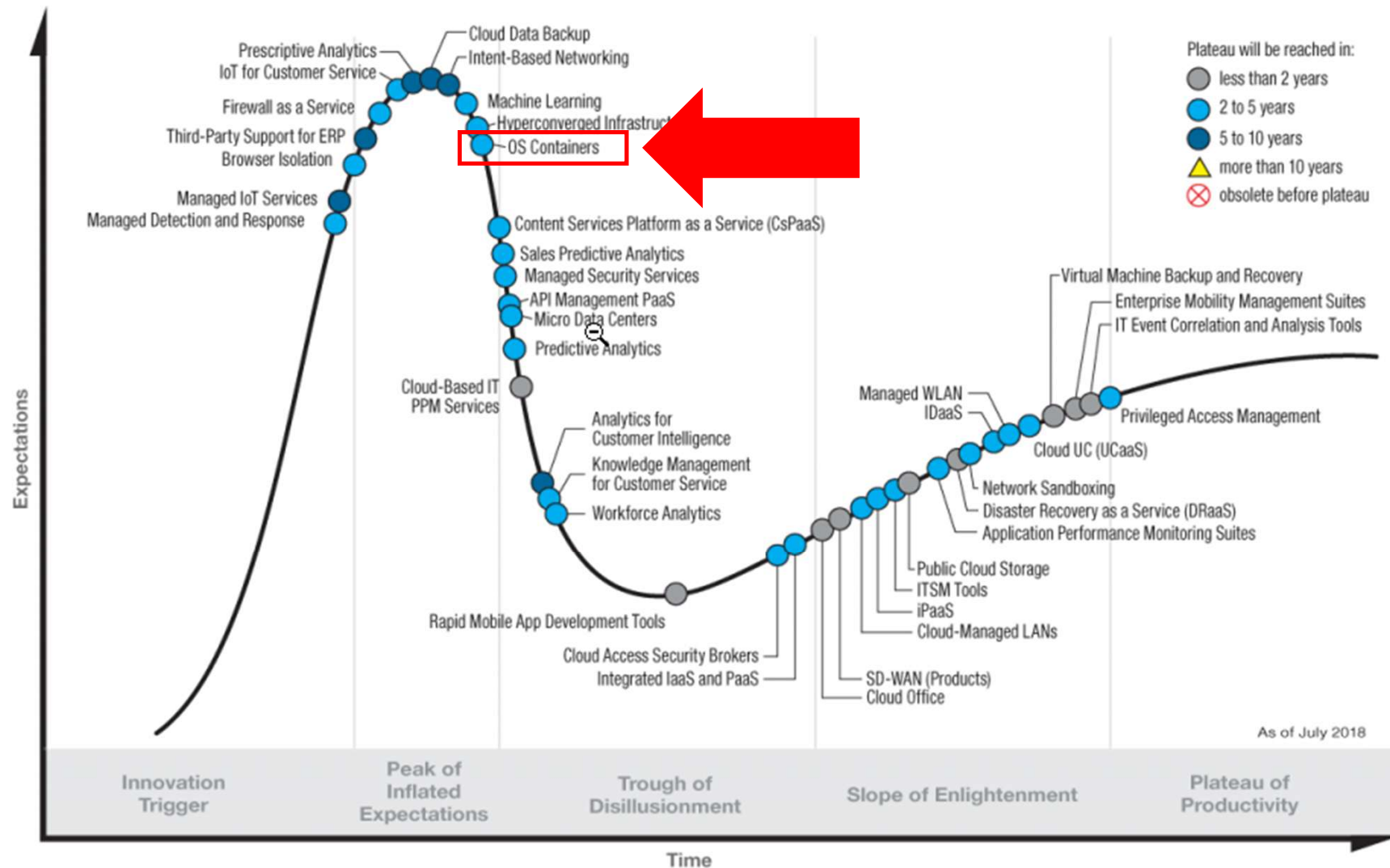


From "Hype Cycle for Infrastructure Strategies, 2017," 27 July 2017 (G00313602)

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Gartner

Hype Cycle for Midsize Enterprises, 2018



<https://www.gartner.com/smarterwithgartner/5-strategic-technologies-on-the-gartner-hype-cycle-for-midsize-enterprises-2018/>

By 2018, more than 50% of new workloads will be deployed into containers ~~in at least one stage of the application life cycle.~~



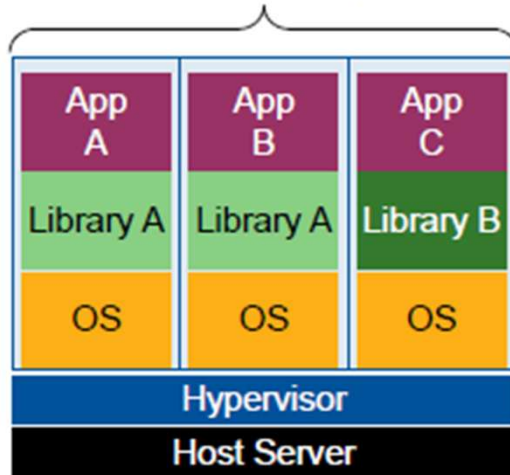
Contenedores

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2da Ola de Virtualización

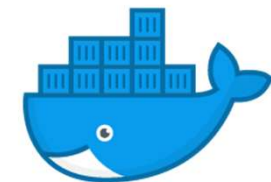
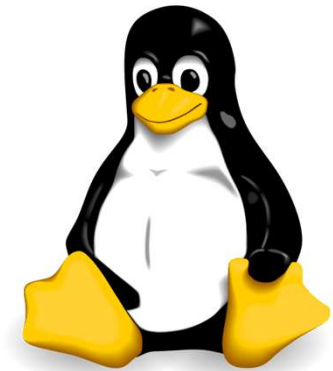
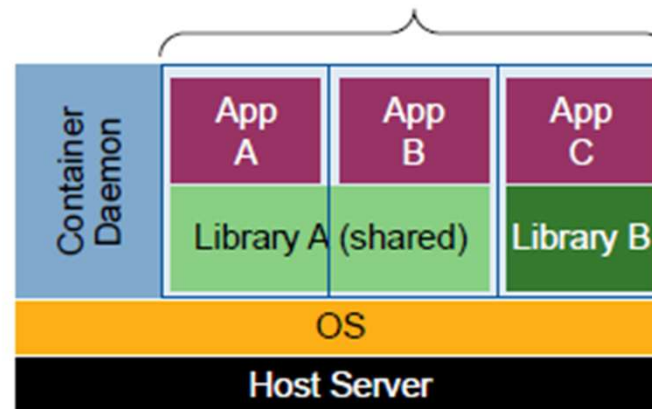
Virtual Machines:

- Self-Contained
- Isolated/Secure
- Administration-Focused



Containers:

- Lightweight
- High-Density
- Developer-Focused



docker

Complementariedad

VMs

Infrastructure administrator

Hardware consolidation,
capacity management

Monolithic, vertical, rapid
provisioning

Hypervisor-dependent

User

Benefit

Ideal Applications

Security

Containers

Developer

Ease of development,
development libraries

Microservices, horizontal,
volatile scale or change

OS-dependent

Beneficios (Tiempo)

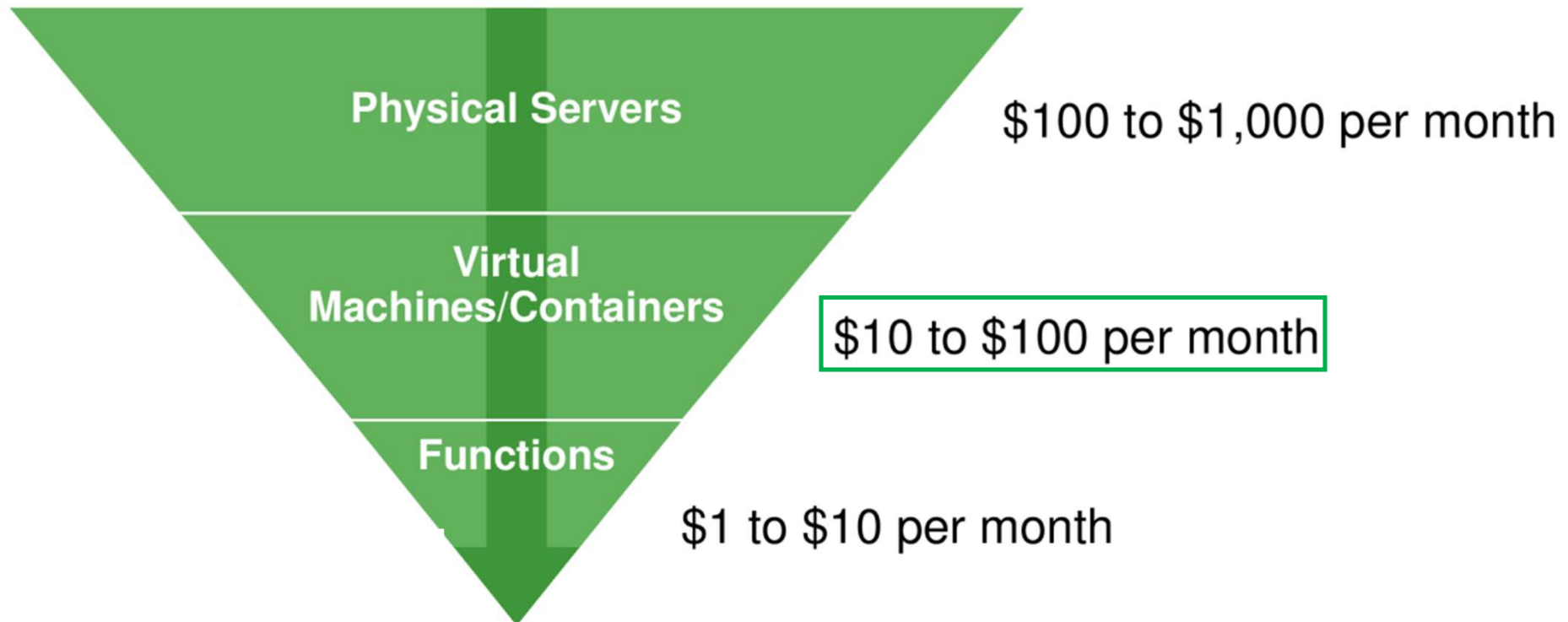
Base Technology Enablers: Agility and Ease of Use @ the core of change

Drivers: Workloads (mostly new), ease of use (both manageability and developer).

Technology	Time to Deploy	Instance Life Span
Physical Servers	One to Three Months	A Few Years
Virtual Machines	Minutes to Days	A Few Weeks
Containers	Milliseconds to Seconds	A Few Days
Serverless	Milliseconds	A Few Seconds

Don't get locked up in only one technology — agility requires a range

Beneficios (Costos)



Recomendaciones

- ✓ Use containers now for **development/test** (any application architecture).
- ✓ Use containers now for **greenfield microservices** architecture/web-scale apps in public cloud.
- ✓ **Evolve IaaS cloud** architectures toward containers as the primary unit of service.
- ✓ Evaluate supporting **containers in production** for traditional applications, but monitor the improving maturity of the container ecosystem.
- ✓ **Complement containers with VMs** for now for mature management, proven isolation/security and separation of concerns.

Contenedores

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- 3. ¿Cuál es la estrategia?**
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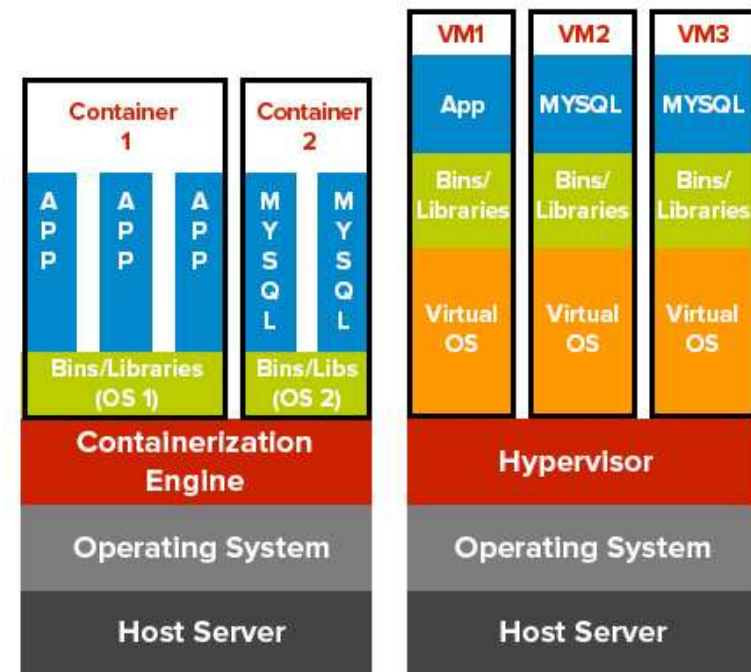
Containers



Transformar

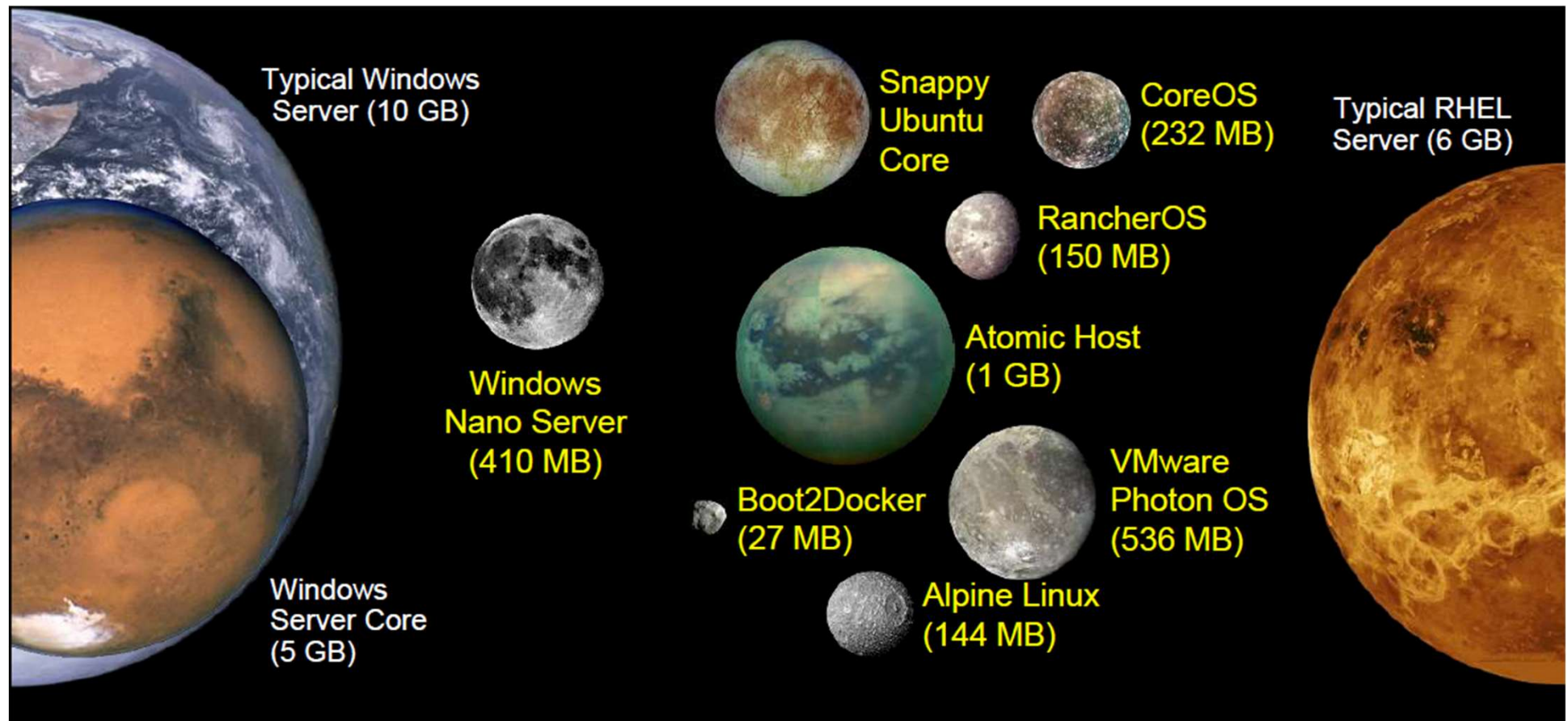
- Desarrollo (DEV/QA)
 - Servidores
 - Estaciones de Trabajo
- Producción (PRD)
 - Aplicaciones Web
 - Micro-servicios (API)

Containerization vs Virtualization

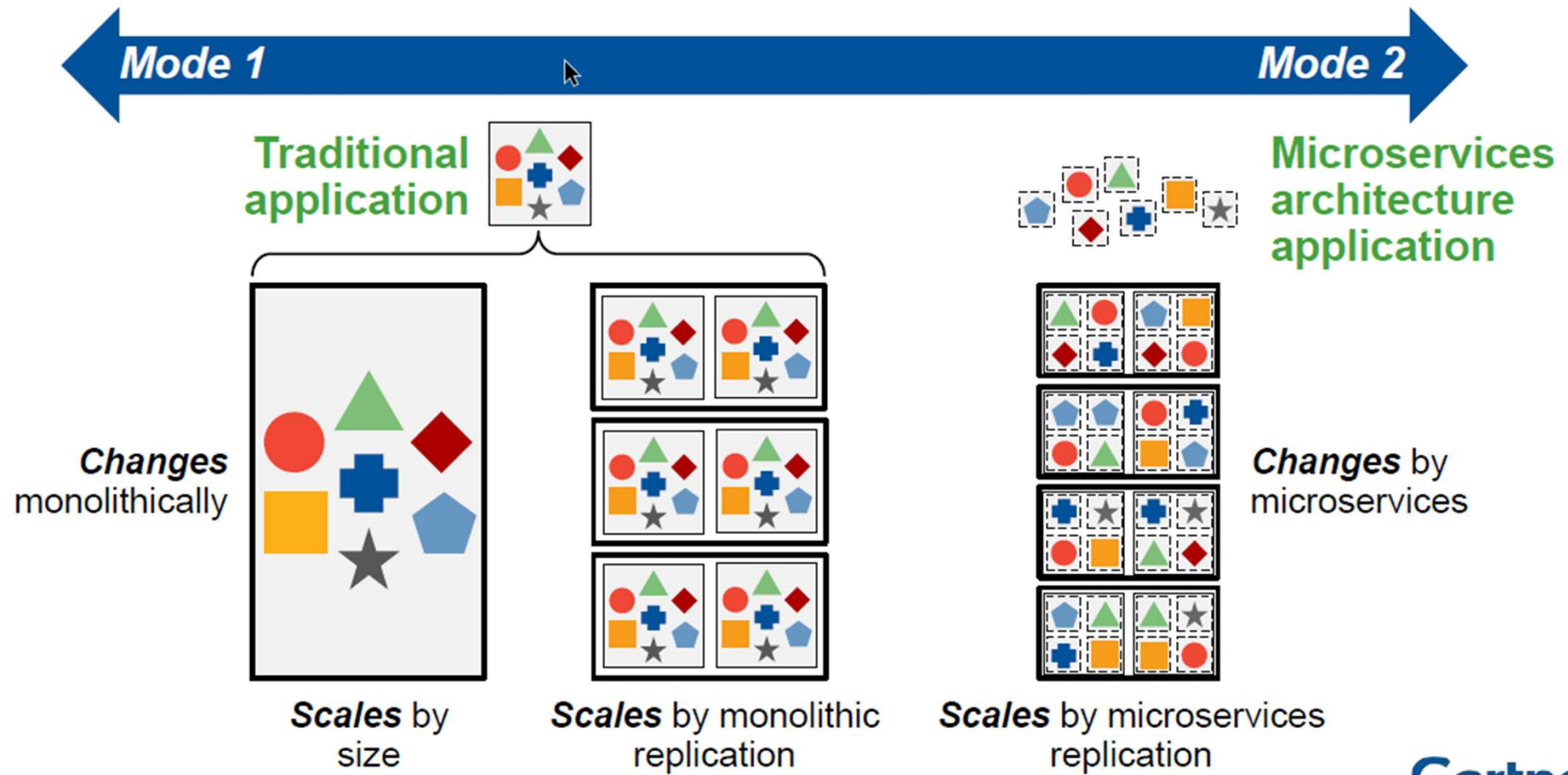


<https://www.smartfile.com/blog/wp-content/uploads/2015/10/containerization-vs-virtualization.jpg>

Optimizar

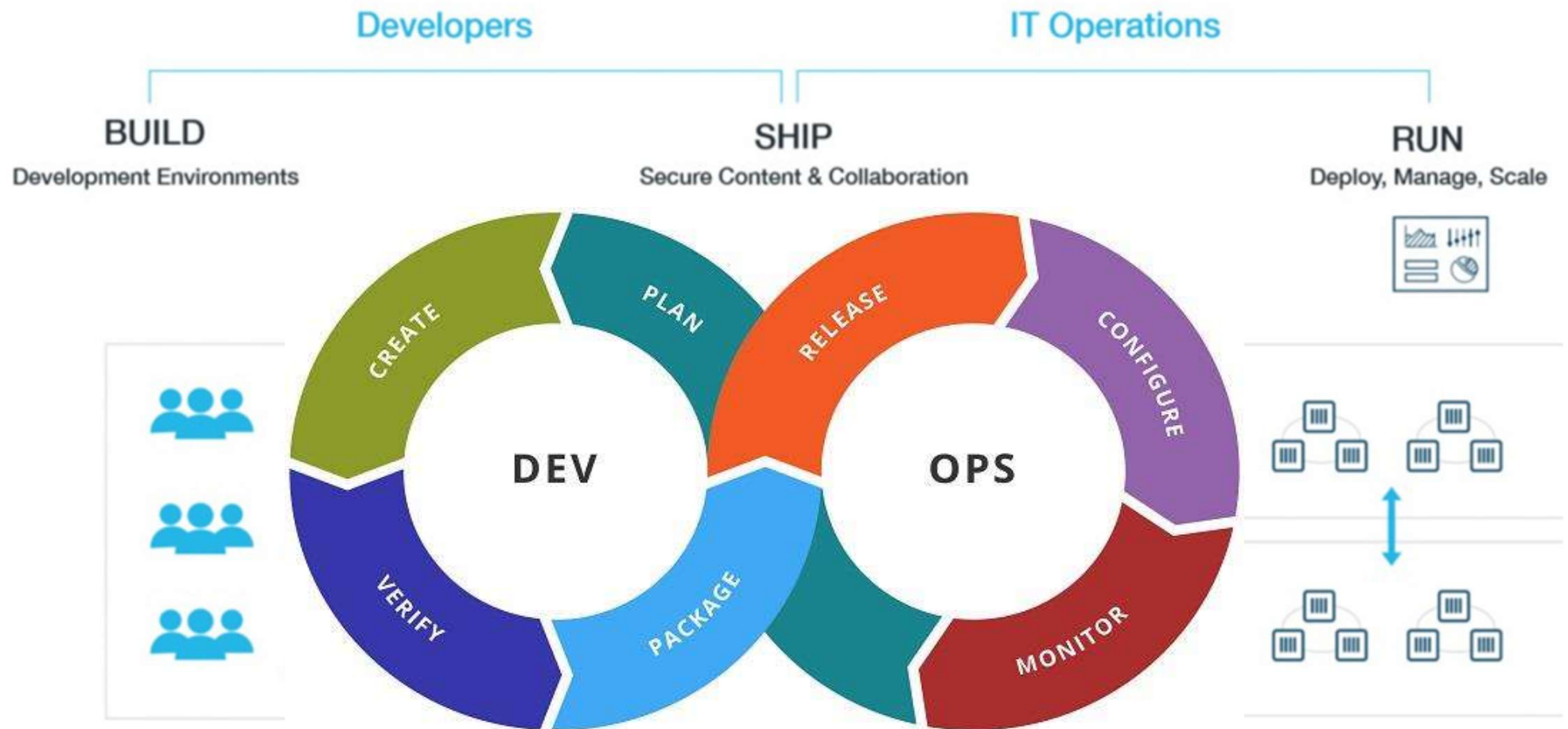


Arquitectar



Agilizar

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Contenedores

1. ¿Por qué tanto escándalo?
2. ¿Qué son los contenedores?
3. ¿Dónde y para qué usarlos?
4. **¿Cuál es la arquitectura?**

Docker Momentum



450+

Docker EE
commercial
customers



37B

Container
downloads



15K

Job listings on
LinkedIn



3.5M

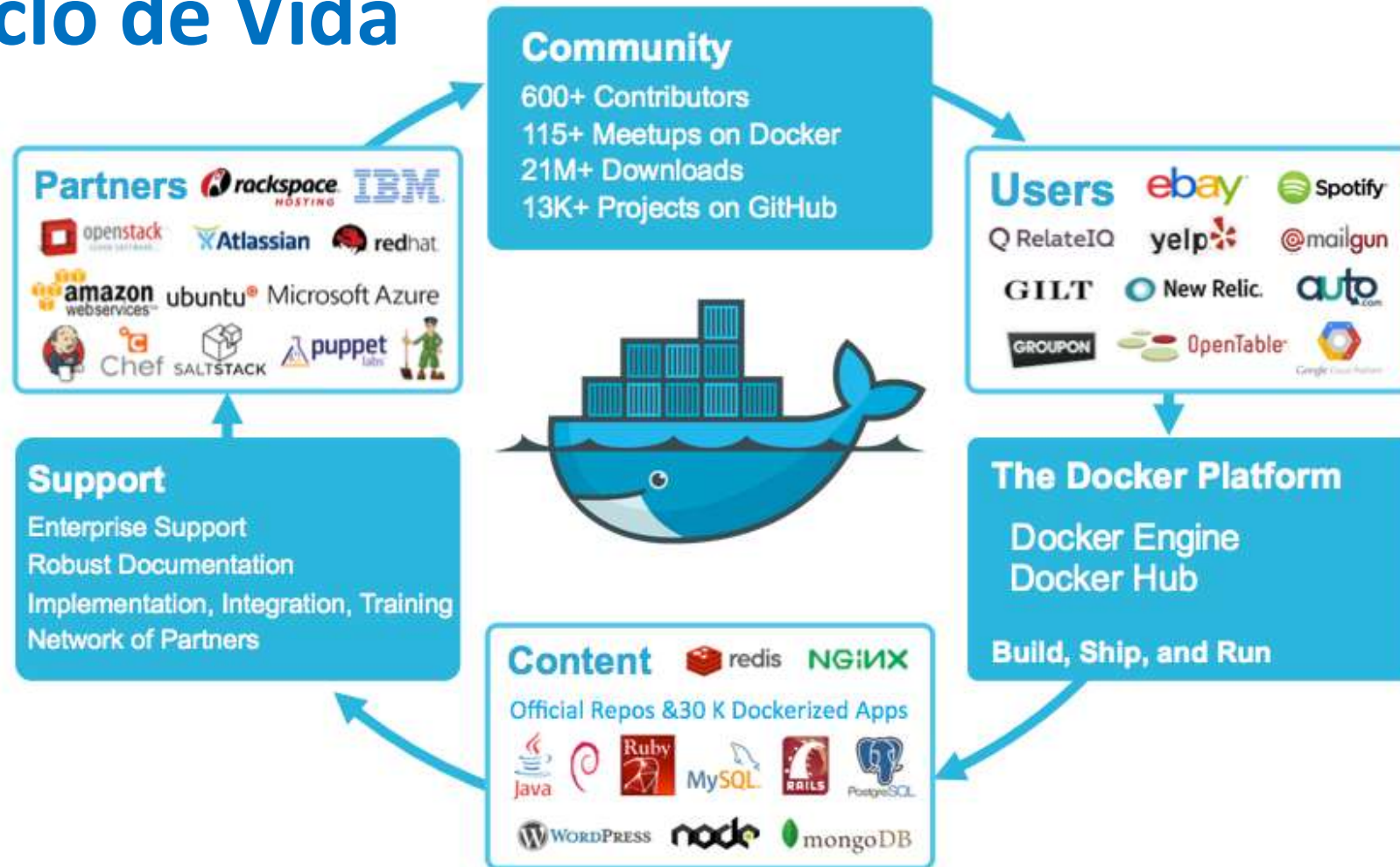
Dockerized
apps



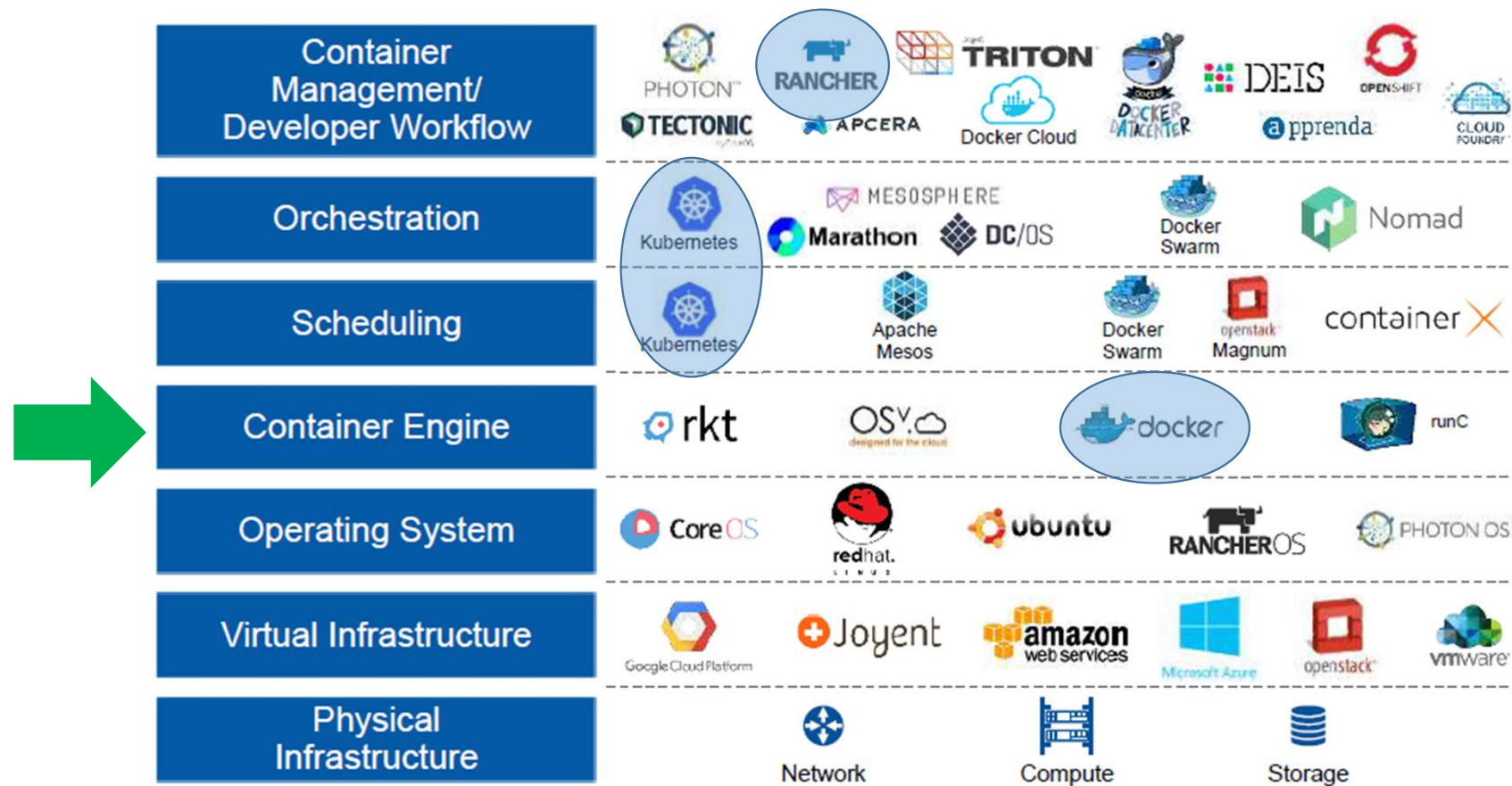
200+

Active Docker
user groups

Ciclo de Vida



Ecosistema

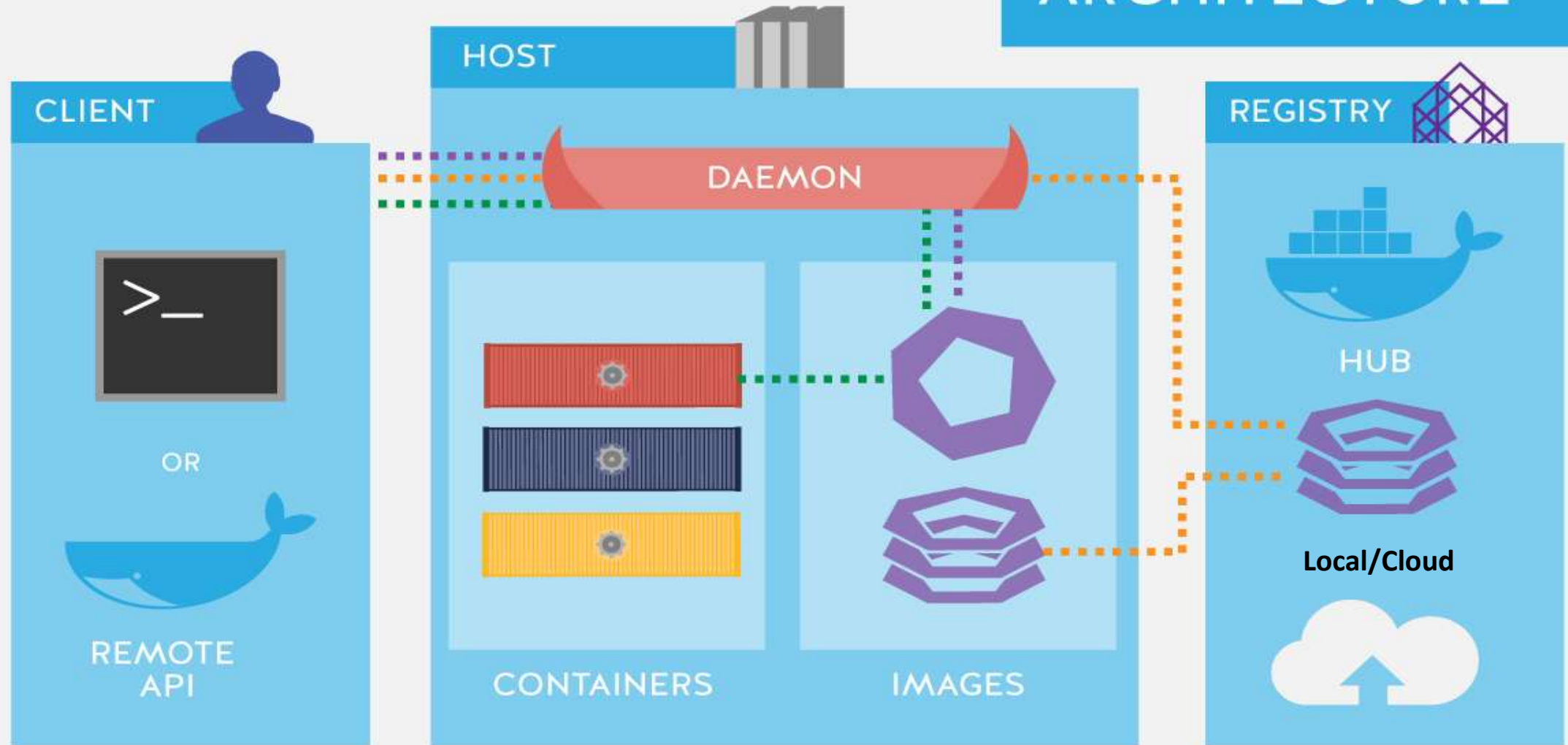


BUILD

PULL

RUN

DOCKER ARCHITECTURE



Ecosistema (Mega-Nubes)



Amazon EKS

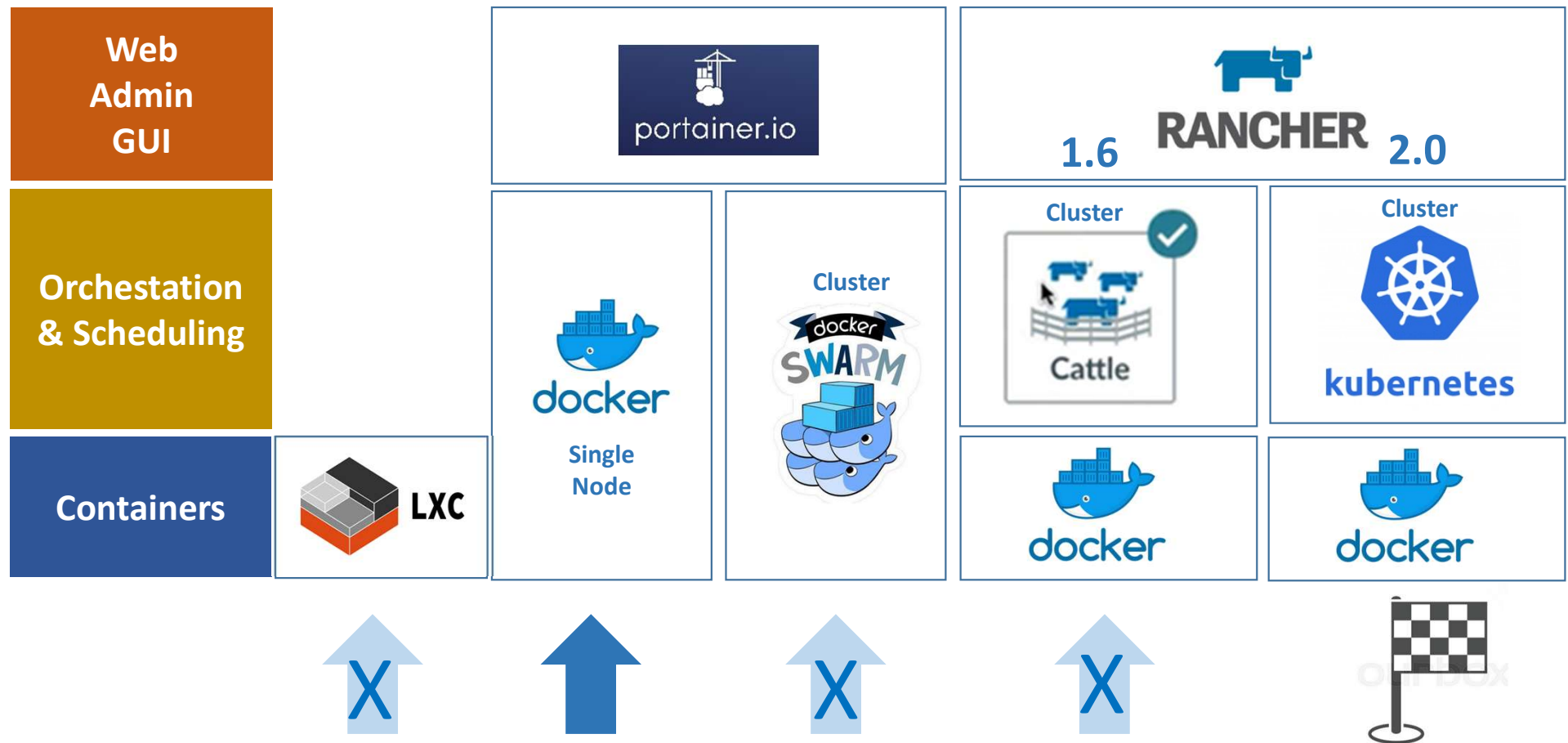


kubernetes



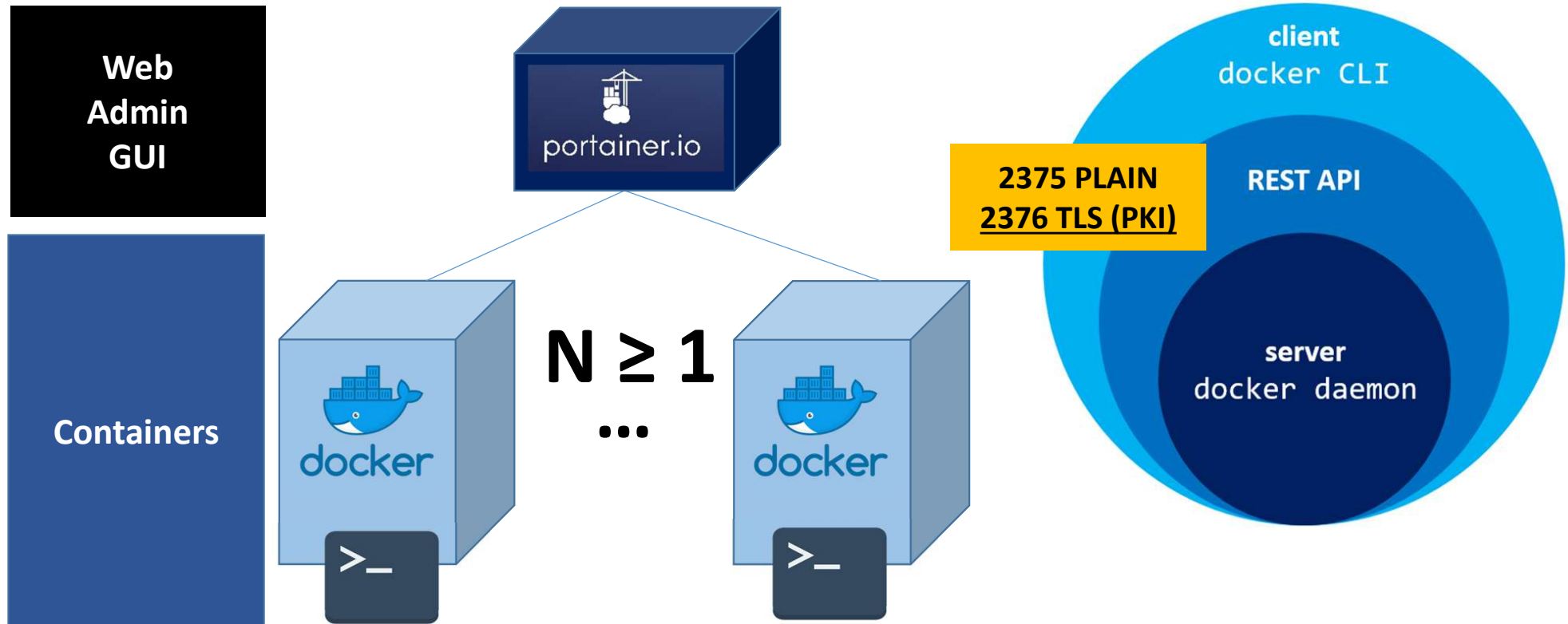
docker

Ecosistema (Test Drive)



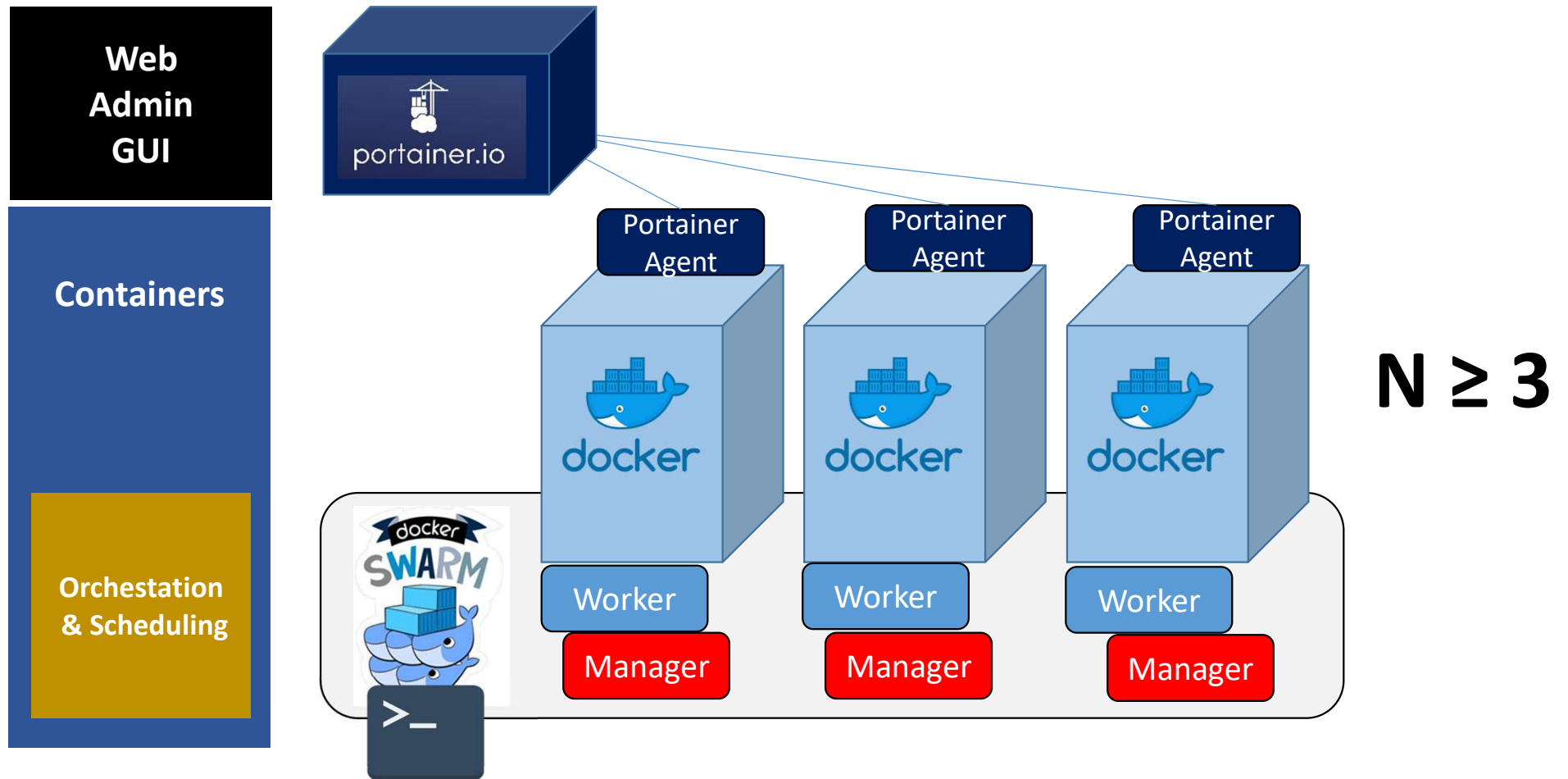
Arquitectura (Docker::Standalone)

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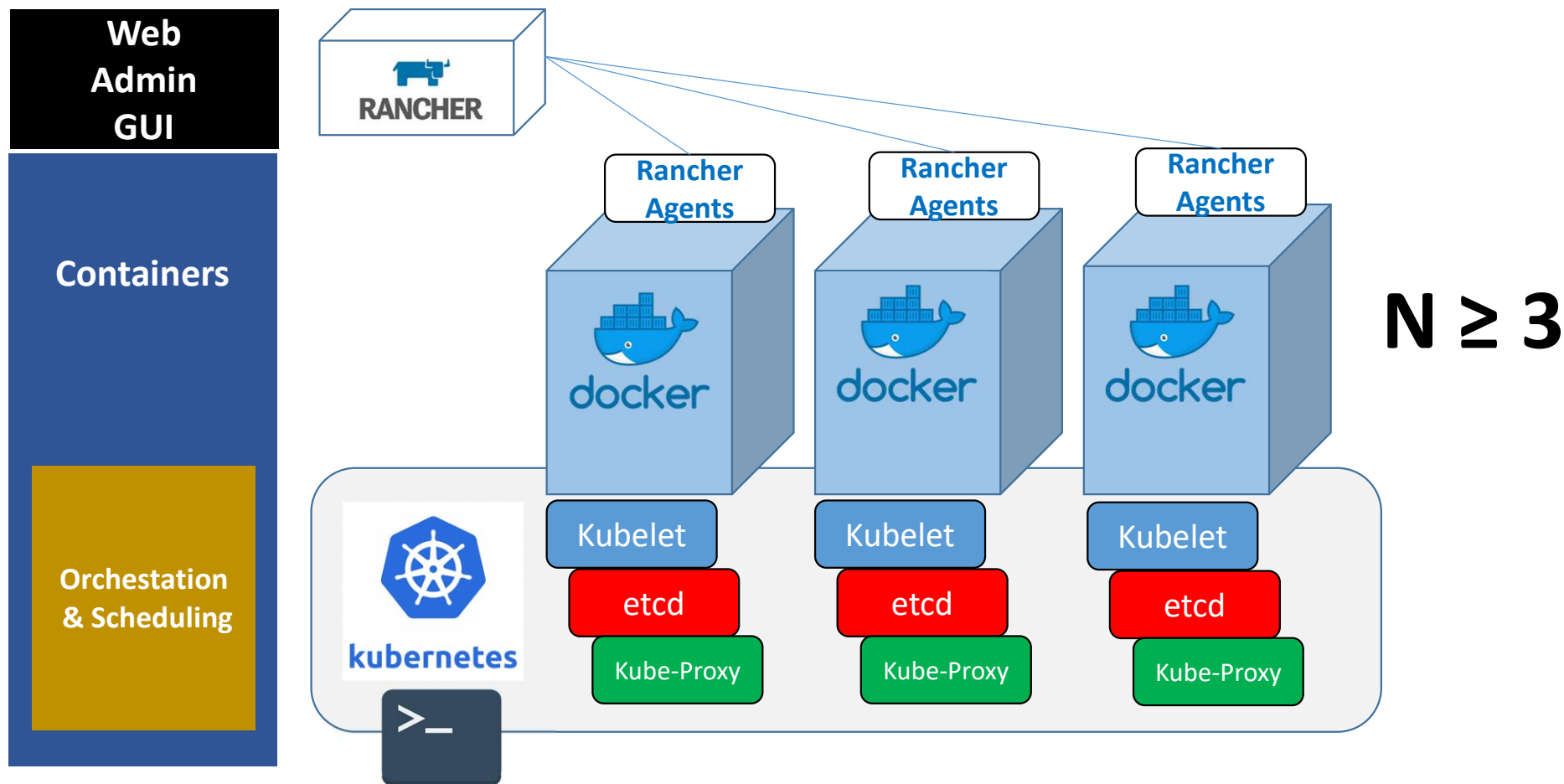
Arquitectura (Docker::Swarm)

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Arquitectura (Docker::Kubernetes)

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Arquitectura (Seguridad)

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12 Security Capabilities of the Gartner Adaptive Security Architecture

